

## Project Design Phase-II

### Restrict Record Access Based on Field Value

|               |                       |
|---------------|-----------------------|
| Date          | 29 September 2026     |
| Team ID       | SWTID-2026-6990       |
| Project Name  | Script-Controlled ACL |
| Maximum Marks | 4 Marks               |

#### Technical Architecture:

The architecture describes how ServiceNow Users and Roles work with a Script-Controlled ACL to restrict record access based on a selected field value.

Project Architecture: User / Role Assignment → ServiceNow Application → ACL Evaluation → Field Value Check → Allow or Deny Record Access

Security focus: role-based access with script-controlled field-level conditions; no external API or machine-learning model is required for this milestone.

## ARCHITECTURE FLOW

Users / Roles



ServiceNow Table



Script-Controlled ACL

ACL checks the required field value and user authorization → Access Granted / Access Denied

**Table-1 : Components & Technologies:**

| S.No | Component                       | Description                                                                                        | Technology                                  |
|------|---------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------|
| 1    | User Interface                  | How administrators and developers interact with the ServiceNow instance and test record access.    | ServiceNow Platform UI / Web UI             |
| 2    | Application Logic-1             | User and role setup required for the access-control requirement.                                   | ServiceNow Users, Roles and Groups          |
| 3    | Application Logic-2             | ACL evaluates whether the logged-in user satisfies the required access condition.                  | ServiceNow ACL Engine                       |
| 4    | Application Logic-3             | Script checks the selected record field value and returns access permission.                       | JavaScript ACL Script                       |
| 5    | Database                        | Stores users, roles, groups, records and field values used by the ACL.                             | ServiceNow Platform Database                |
| 6    | Cloud Database                  | Cloud-hosted data services used by the ServiceNow platform when the instance is deployed in cloud. | ServiceNow Cloud Database                   |
| 7    | File Storage                    | Not required for this ACL milestone; configuration is stored in the ServiceNow instance.           | ServiceNow Configuration / Platform Storage |
| 8    | External API-1                  | No external API is required for the ACL access-control logic.                                      | Not Applicable                              |
| 9    | External API-2                  | No external API is required for the ACL access-control logic.                                      | Not Applicable                              |
| 10   | Machine Learning Model          | No machine-learning model is required because access is rule/script based.                         | Not Applicable                              |
| 11   | Infrastructure (Server / Cloud) | ServiceNow instance hosts the application, ACL rules, users, roles and data.                       | ServiceNow Cloud Instance                   |

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                                                                                                                                              | Technology                                            |
|------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 1    | Open-Source Frameworks   | No separate open-source application framework is required for this milestone; the solution is implemented within ServiceNow configuration and scripting. | ServiceNow scripting / JavaScript                     |
| 2    | Security Implementations | Role-based and script-controlled access is applied to records. ACL logic checks the required field value before granting access.                         | ServiceNow ACLs, Roles, Groups, JavaScript conditions |
| 3    | Scalable Architecture    | Access-control logic is centralized in the ServiceNow ACL layer, allowing the same rule to be evaluated consistently across records and users.           | ServiceNow ACL Engine / Platform Architecture         |
| 4    | Availability             | The ACL is executed within the ServiceNow instance, so access decisions are available whenever the platform is available.                                | ServiceNow Cloud Platform                             |
| 5    | Performance              | Use simple field comparisons and role checks in ACL scripts; avoid unnecessary database calls or complex loops to keep access evaluation lightweight.    | JavaScript ACL Script / ServiceNow Platform           |

### References:

ServiceNow ACL and Access Control documentation

ServiceNow Users, Roles and Groups configuration

ServiceNow JavaScript scripting for ACL conditions

ServiceNow platform security concepts

Project requirement: Restrict Record Access Based on Field Value